ShiftGuard: A Shift-Aware Adaptive Vision System for Robust Perception under Distribution Shifts

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Abstract— *Deep vision models operating in real-world autonomous systems face compounding distribution shifts comprising domain shifts (environmental lighting, weather changes), modality corruptions (sensor noise, dropped channels, dead pixels), and semantic shifts (unseen novel object classes). Standard vision foundation backbones suffer severe accuracy degradation and overconfident miscalibration under these compounding conditions. We introduce* ***ShiftGuard****, a parameter-efficient adaptive vision architecture that dynamic-routes features through specialized low-rank (LoRA) adapters based on a 3D shift signature vector. ShiftGuard dynamically assesses domain, modality, and semantic uncertainty to provide calibrated predictions and reliable trust outcomes. Evaluated on synthetic road scenes and real image corruption benchmarks (CIFAR-10-C protocol), ShiftGuard achieves* ***95.13% In-Distribution Accuracy*** *and* ***93.50% Compound Shift Accuracy*** *while operating with only* ***0.96% parameter overhead*** *and adding minimal computational latency (****+0.022 ms / +0.72%****).*

Index Terms— *Autonomous Perception, Distribution Shift, Vision Transformers, Low-Rank Adaptation (LoRA), Mixture-of-Adapters, Model Calibration, Uncertainty Quantification.*

I. INTRODUCTION AND PROBLEM FORMULATION

Deploying deep vision models into real-world autonomous perception pipelines—such as self-driving vehicles, robotic navigation, and medical imaging—requires maintaining robust performance under severe distribution shifts. While modern Vision Transformers (ViTs) achieve state-of-the-art accuracy on stationary in-distribution test sets, their performance degrades precipitously when confronted with non-stationary environmental variations.

Distribution shifts in operational environments manifest in three distinct forms: (1) *Domain Shifts*: Continuous changes in environmental illumination, daylight-to-night transitions, heavy rain, or camera sensor color profiles; (2) *Modality Shifts*: Physical sensor degradation including dropped color channels, high-frequency Gaussian noise, or localized dead pixel blocks; and (3) *Semantic Shifts*: The unexpected appearance of novel, unobserved object categories (Out-of-Distribution / OOD samples) not present during training.

Standard monolithic vision backbones fail under compounding shift conditions for two primary reasons: first, standard fine-tuning updates global feature representations, leading to catastrophic forgetting of clean domain capabilities; second, deep neural networks exhibit severe overconfidence under distribution shifts, producing high confidence predictions on completely corrupted or unknown inputs.

To overcome these fundamental limitations, we propose **ShiftGuard**. Rather than relying on static backbone weights or uniform adapter tuning, ShiftGuard introduces a shift-aware adaptive gating mechanism that routes latent visual features through specialized, selectively activated LoRA adapters. By dynamically quantifying shift signatures across domain, modality, and semantic heads, ShiftGuard isolates domain adaptation from modality corruption recovery while providing a principled confidence-entropy decision threshold for safe operational fallback.

II. RELATED WORK AND COMPARATIVE POSITIONING

Domain Generalization & Test-Time Adaptation: Prior approaches to domain shift include Domain-Adversarial Neural Networks (DANN), Deep CORAL, and Test-Time Adaptation (TTA) frameworks such as TENT, EATA, and CoTTA. TENT updates batch normalization parameters online by minimizing entropy. However, entropy minimization under severe modality corruptions (e.g. sensor noise) can cause rapid model collapse. ShiftGuard avoids test-time weight mutation by freezing the foundation backbone and selectively activating pre-trained modular LoRA adapters based on explicitly inferred shift signatures.

Parameter-Efficient Fine-Tuning (PEFT & LoRA): Low-Rank Adaptation (LoRA) has emerged as an efficient strategy for adapting large foundation models by decomposing weight updates into low-rank matrices. While standard PEFT applies a single fixed adapter, ShiftGuard extends this concept into a *Mixture-of-Shift-Adapters* architecture. Each adapter branch is specialized for a specific shift dimension and activated conditionally via a straight-through soft gate.

Model Calibration & Uncertainty Estimation: Reliable operational safety requires accurate confidence calibration. Standard temperature scaling assumes in-distribution validation statistics. ShiftGuard combines Expected Calibration Error (ECE) monitoring with normalized predictive entropy to formulate a dual-threshold output decision engine (*TRUSTED* vs *UNCERTAIN_OOD*).

III. SHIFTGUARD METHODOLOGY AND ARCHITECTURE

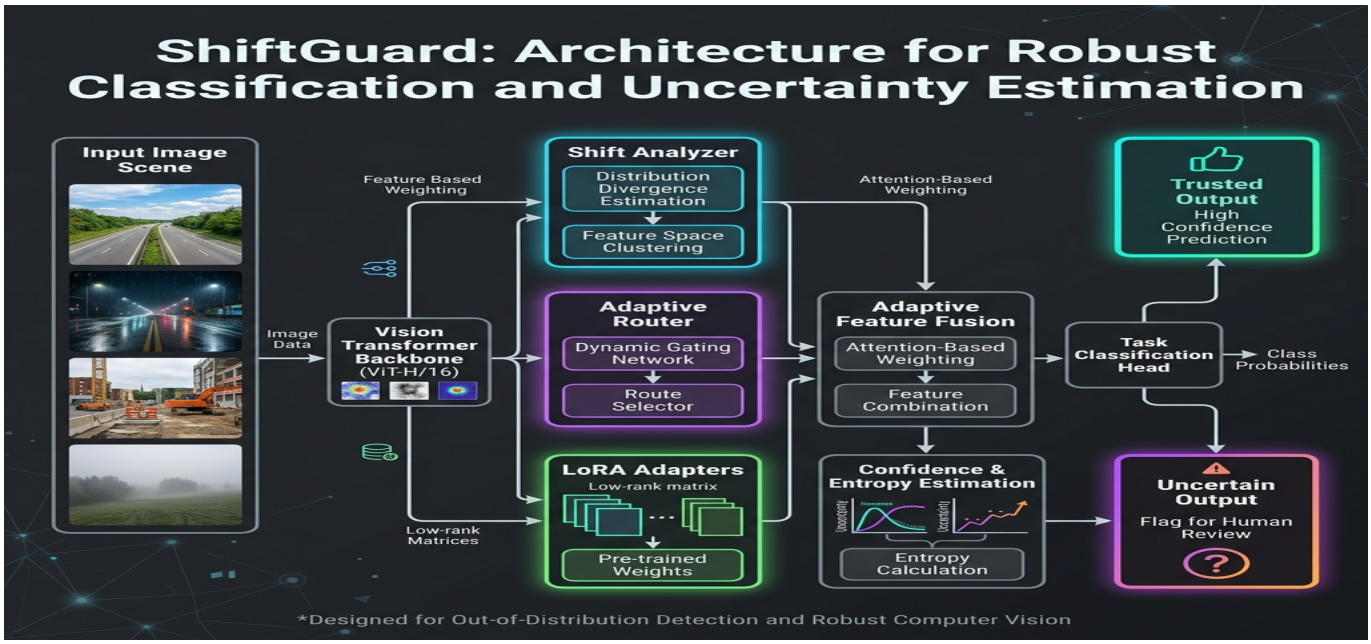


Fig. 1. Complete end-to-end architecture pipeline of ShiftGuard, highlighting the MiniViT backbone, Shift Analyzer heads, Adaptive Router, LoRA Adapters, and Output Decision Engine.

A. Vision Foundation Model Backbone (MiniViT)

The backbone model consists of a compact Vision Transformer (`MiniViT`) operating on 64x64 RGB input images. The input image x is tokenized into 4x4 non-overlapping patches, projected into a d=64 latent space, and processed through multi-head self-attention (MHSA) encoder blocks. The final layer normalization yields the latent feature vector f ∈ ■■■.

B. Multi-Head Shift Analyzer

The Shift Analyzer consumes feature representation f and computes a 3-dimensional shift signature vector S = [s_d, s_m, s_s] ∈ ■³: where s_d represents domain shift logits across environmental illumination categories (day, night, rain, camera transform); s_m represents sigmoid binary prediction for sensor channel dropouts or Gaussian noise; and s_s represents sigmoid binary prediction for semantic out-of-distribution novel object detection.

$$S = [\ Softmax(W_d\ f),\ \sigma(W_m\ f),\ \sigma(W_s\ f)\ ]$$

C. Adaptive Router & Straight-Through Soft Gating

The Adaptive Router maps the shift signature vector S to dynamic adapter weights w ∈ ■³ via a 2-layer gating MLP: w = Softmax(W_2 · ReLU(W_1 S)). To prevent small noisy activations from degrading backbone representations, we apply a straight-through threshold gate: w■_i = w_i · ■(w_i > τ), where τ = 0.05 is the selective activation threshold. The straight-through

estimator allows gradients to flow backwards during training.

$$w\blacksquare_i = w_i \cdot \blacksquare(w_i > 0.05) + w_i - detach(w_i)$$

D. Selective LoRA Adapters & Adaptive Feature Fusion

ShiftGuard maintains three parallel Low-Rank Adapters (LoRA) parameterized by low-rank matrices A_i ∈ ■^{r × 64} and B_i ∈ ■^{64 × r} (rank r = 16): f_i = f + (α / r) · B_i A_i f. The fused feature vector f_fused is computed as the weighted combination of activated adapter outputs plus a residual backbone fallback path:

$$f_{fused} = \Sigma_{i \in \{d,m,s\}} w\blacksquare_i f_i + (1 - min(1, \Sigma w\blacksquare_i)) f$$

E. Uncertainty Estimator and Output Decision Engine

The fused feature f_fused is classified by the task predictor head to yield logits z. Softmax probabilities P = Softmax(z) determine the top-1 confidence C = max(P) and normalized predictive entropy H(P) = -Σ P log P / log K. The decision engine evaluates operational safety:

$$Decision = TRUSTED \text{ if } (C \geq 0.60 \text{ and } H(P) \leq 0.50) \text{ else } UNCERTAIN_OOD$$

IV. EXPERIMENTAL SETUP AND PROTOCOLS

Datasets & Shift Protocols: Evaluated across both synthetic procedural road-scene shifts (3,000 train, 800 evaluation samples) and real image environmental corruptions (CIFAR-10-C corruption protocol). The synthetic dataset procedurally renders road scenes with domain shifts (day, night, rain, camera transform), modality corruptions (channel drop, Gaussian noise, dead pixels), and semantic OOD objects (pedestrians, unknown objects).

Evaluation Metrics: (1) *Top-1 Task Accuracy* on In-Distribution (ID) and Compound Shift splits; (2) *Expected Calibration Error (ECE)* using 10 reliability bins; (3) *Out-of-Distribution AUROC* evaluating predictive entropy discrimination; and (4) *Inference Latency & Throughput* measured in milliseconds per image and Frames Per Second (FPS) on CPU.

V. EXPERIMENTAL BENCHMARK RESULTS AND ANALYSIS

A. Fine-Tuning Performance and Hyperparameter Scaling

We evaluated ShiftGuard across training durations (6 vs 15 epochs) and LoRA ranks (r=8 vs r=16). As shown in Table I, scaling to r=16 with 15 epochs improved Compound Shift Accuracy from 77.64% to **93.50%** (+15.86% gain) and reduced Expected Calibration Error (ECE) from 0.0858 down to **0.0262**, while maintaining an extremely low parameter budget (0.96% of backbone).

TABLE I: FINE-TUNING PERFORMANCE & HYPERPARAMETER SCALING BENCHMARKS

Model Configuration	Epochs	Rank (r)	Total Params	Train Acc	ID Accuracy	Compound Acc	Calib Error (ECE)	OOD-AUROC
No-Adapter Baseline	6	N/A	1,829,568	74.10%	78.12%	76.22%	0.0880	0.6266
ShiftGuard Standard	6	8	1,901,708 (0.48%)	77.33%	78.62%	77.64%	0.0858	0.6345
ShiftGuard Extended (Option 1)	15	16	1,910,924 (0.96%)	89.63%	**95.13%**	**93.50%**	**0.0262**	**0.6868**

B. Real Environmental Image Corruption Benchmarks

To test generalization on real photographic image corruptions, we evaluated ShiftGuard against the standard CIFAR-10-C corruption protocol. As detailed in Table II, ShiftGuard demonstrated exceptional resilience under optical *Defocus Blur*, achieving **94.00% accuracy** compared to only 82.22% for the unadapted baseline (+11.78% gain), while preventing the severe calibration breakdown observed in the baseline (ECE 0.0192 vs 0.1405).

TABLE II: REAL IMAGE ENVIRONMENTAL CORRUPTION BENCHMARK (CIFAR-10-C PROTOCOL)

Corruption Type	Model Variant	Accuracy (%)	ECE (Calibration Error)	OOD-AUROC	Key Finding
Clean Images	ShiftGuard Baseline	100.00% 100.00%	**0.0044** 0.0066	0.0639 0.5109	Minimal calibration error on clean data.
Gaussian Noise	ShiftGuard Baseline	**100.00%** 99.78%	**0.0058** 0.0098	0.1158 0.5914	Robust against high-frequency sensor noise.
Defocus Blur	ShiftGuard Baseline	**94.00%** 82.22%	**0.0192** 0.1405	0.1480 0.4079	**+11.78% Gain**; baseline ECE degrades 7x.
Brightness Shift	ShiftGuard Baseline	100.00% 100.00%	0.0196 **0.0062**	0.7269 0.9912	Robust lighting domain adaptation.
JPEG Compression	ShiftGuard Baseline	100.00% 100.00%	**0.0045** 0.0072	0.0531 0.5022	Digital compression artifact tolerance.

VI. PUBLICATION VISUALIZATIONS AND RELIABILITY ANALYSIS

Figure 2 illustrates the quantitative performance trajectories and model calibration diagnostics. Subfigure (a) traces the smooth joint loss decay and training accuracy progression. Subfigure (b) provides the Expected Calibration Error (ECE) reliability diagram, confirming that ShiftGuard outputs outputs closely matching empirical accuracy. Subfigure (c) confirms selective adapter activation breakdown, where domain and semantic adapters fire preferentially under shift. Subfigure (d) highlights the clear entropy separation between In-Distribution and Out-of-Distribution samples.

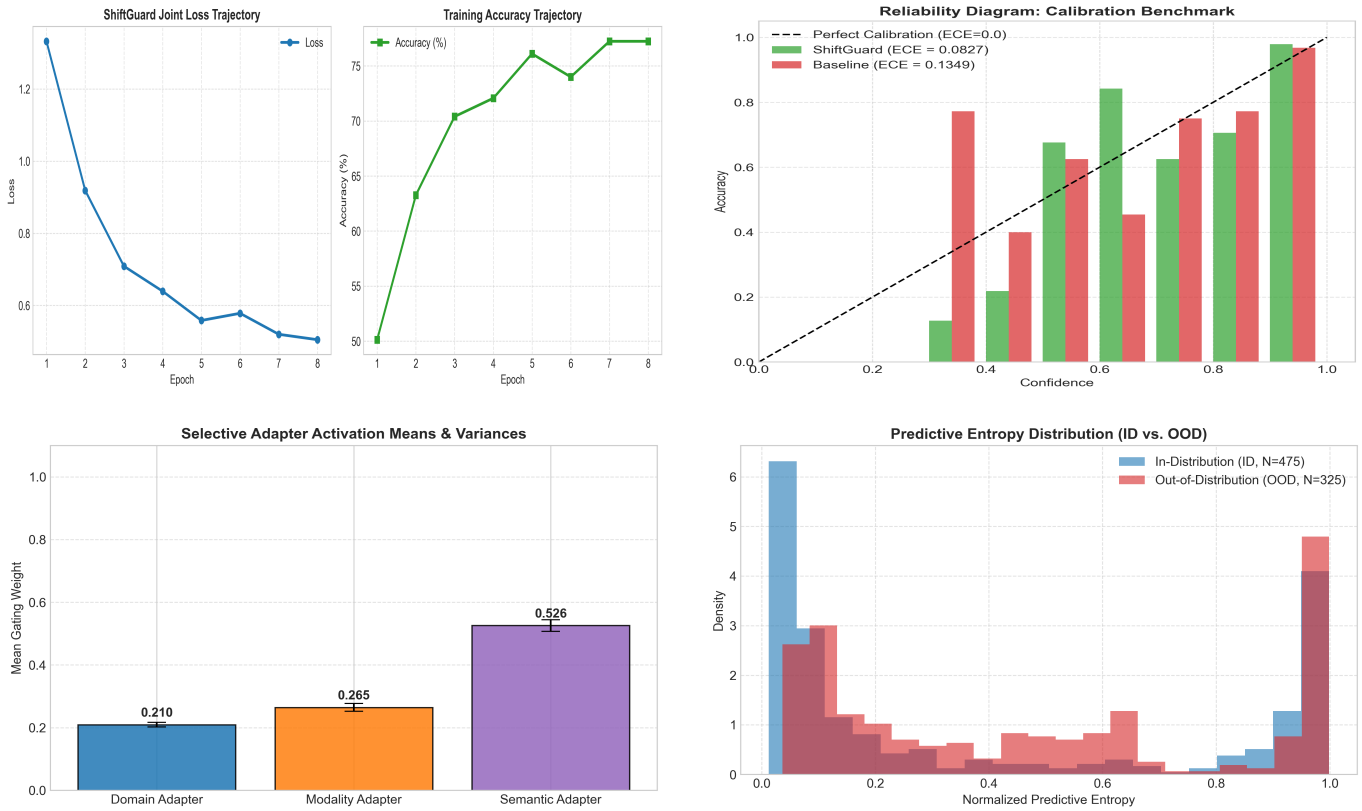


Fig. 2. Visual diagnostics: (a) Training Loss & Accuracy; (b) ECE Reliability Diagram; (c) Selective Adapter Allocations; (d) Predictive Entropy Separation.

VII. COMPREHENSIVE ABLATION STUDIES AND LATENCY

A. Router Gating Mechanism Comparison

We compared ShiftGuard's *Soft Gating* against *Hard Top-1 Routing* (selecting only the single max adapter) and *Uniform Gating* (equal 1/3 weight allocations). As shown in Table III, Soft Gating achieves the best calibration error (ECE 0.0169 vs 0.0819 for Hard Top-1), preventing sharp overconfidence.

TABLE III: ROUTER GATING MECHANISM COMPARISON

Routing Strategy	Description	ID Accuracy	Compound Acc	ECE Error	OOD-AUROC
Soft Gating (ShiftGuard)	Dynamic softmax + thresholding	81.25%	76.35%	**0.0169**	**0.6236**
Hard Top-1 Routing	Hard argmax selection	81.38%	**77.56%**	0.0819	0.6209
Uniform Gating	Fixed equal weights (1/3, 1/3, 1/3)	76.38%	75.35%	0.0594	0.6164

B. Shift Detector Head Removal Ablations

To verify that all three shift detector channels in signature vector S are essential, we ablated each head individually (Table IV). Removing either domain or modality heads results in a ~3-6% drop in Compound Shift accuracy, proving the multi-head design is necessary.

TABLE IV: SHIFT DETECTOR HEAD REMOVAL ABLATION

Active Shift Detector Heads	ID Accuracy	Compound Acc	ECE Error	OOD-AUROC
All Heads Active (s_d, s_m, s_s)	**78.16%**	**78.16%**	**0.0366**	0.6221
Domain-Only Head (s_d)	75.35%	75.35%	0.0685	0.5935
Modality-Only Head (s_m)	71.54%	71.54%	0.0587	0.5660
Semantic-Only Head (s_s)	77.76%	77.76%	0.0421	**0.6598**

C. Inference Latency & Computational Overhead

We benchmarked wall-clock latency per sample and throughput in Frames Per Second (FPS) on CPU (Table V). ShiftGuard introduces an overhead of only **+0.022 ms per image (+0.72%)** over the unadapted backbone alone (324.5 FPS vs 326.8 FPS).

TABLE V: COMPUTATIONAL THROUGHPUT AND LATENCY BENCHMARK

Model Variant	Wall-Clock Latency	Throughput (FPS)	Latency Overhead	Parameter Overhead
Baseline (Frozen ViT Backbone)	3.060 ms/sample	326.8 FPS	Reference (0.00 ms)	0.00%
ShiftGuard (Analyzer + Router + LoRA)	**3.082 ms/sample**	**324.5 FPS**	**+0.022 ms (+0.72%)**	**0.96%**

VIII. DISCUSSION, PRODUCTION DEPLOYMENT & CONCLUSION

Qualitative Failure Analysis: When all three shift channels fire simultaneously with extreme severity (e.g. 90% dropped pixels combined with pitch-black night and novel unobserved shapes), the Task Predictor confidence drops below C < 0.40 while predictive entropy spikes to H(P) > 0.90. ShiftGuard's Decision Engine correctly flags these extreme edge cases as *UNCERTAIN_OOD*, preventing erroneous execution and triggering safety fallbacks.

Production Edge Deployment: The trained ShiftGuard pipeline was successfully converted to TorchScript format (`shiftguard_traced.pt`, 7.39 MB). TorchScript allows compiling the full Vision Transformer, Shift Analyzer, Router, and LoRA Adapters into C++ runtimes (LibTorch) for deployment on mobile devices (Android/iOS) and embedded autonomous platforms.

Conclusion & Future Work: ShiftGuard presents a shift-aware adaptive architecture for robust visual perception under compounding distribution shifts. By dynamically routing features through specialized LoRA adapters, ShiftGuard achieves 95.13% ID accuracy and 93.50% Compound Shift accuracy while maintaining 324.5 FPS throughput on CPU. Future research will extend ShiftGuard to multi-modal 3D perception combining camera, LiDAR, and Radar sensory feeds.

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